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DBMS ASSIGNMENT

Q1. What is a Database Management System (DBMS)? Why is it Required?
A DBMS is software that allows users to create, store, retrieve, update, and manage data in a structured way.
It acts as an interface b/w the user & the database, ensuring that data is organized, secure & easily accessible.
Example - MySQL, Oracle Database, MongoDB

- Data Security: It control access to data through authentication & authorization mechanism.
- Reduces Data Redundancy: DBMS minimises duplicate data to save storage space & maintain consistency.
- Data Integrity: It ensures accuracy & consistency of data using constraints like Primary key & foreign key.

Q2 Why is DBMS better than the traditional file System?
Ans - less Data redundancy - DBMS reduces duplicate data, while file systems often store repeated copies.
Better data consistency - Changes made in DBMS are ~~effected~~ reflected everywhere, unlike file system update may not
Improved Data Security - DBMS provide controll aces & permissions, where file systems have limited security control.

Q3. What is Data Abstraction?
Ans It is the process of hiding unneccessary details of data & showing only the essential information to users.
It helps reduces complexity & improves data security by restricting access to lower level details.

Physical Level (Internal)	Logical level (Conceptual)	View level (External)
[how data is actually stored in memory (files, indexes, storage structure]	[what data is stored in the database & the relationship b/w data]	[shows only required data to users, hiding the rest of the database details]

Q4. What is 3-Schema Architecture? Why is it important?
Ans The 3 Schema Architecture is a database design model that divides a database system into 3 levels to separate user application from physical database storage. It includes view level, logical level, Physical level
It is important - because it provides data independence, improve security by restricting user access, reduces system complexity.

Q5. What is Data independence? Why is it needed
It is the ability to modify the structure of a database at 1 level without affecting the structure at the next higher level.
It is needed because it reduces system complexity, makes database maintanance easier, allows future modification without rewriting application programs & improves flexibility

Q. What is an Entity Relationship (ER) Model?

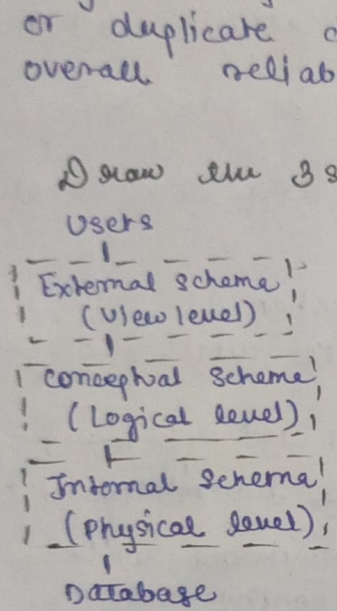
Ans The ER model is a high level data model used to design & map the structure of a database.
It describes data in terms of entities (objects like Student), attributes (properties like name) & relationship (connection b/w entities).

Q1. What are integrity constraints? Why are they important in DBMS?

Ans Integrity constraints are rules applied to a database to ensure the data stored is accurate, valid, & consistent. They include rules like Primary key, Foreign key, Not Null & UNIQUE constraints.
They're imp because they maintain data accuracy, prevent invalid or duplicate data, ensure consistency b/w related tables & improve overall reliability of database system.

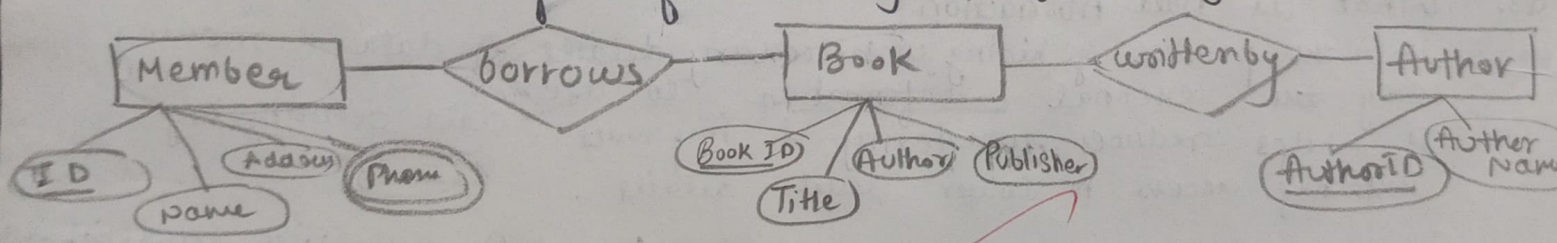
SECTION B

Q1 Draw the 3 schema architecture & explain each level.



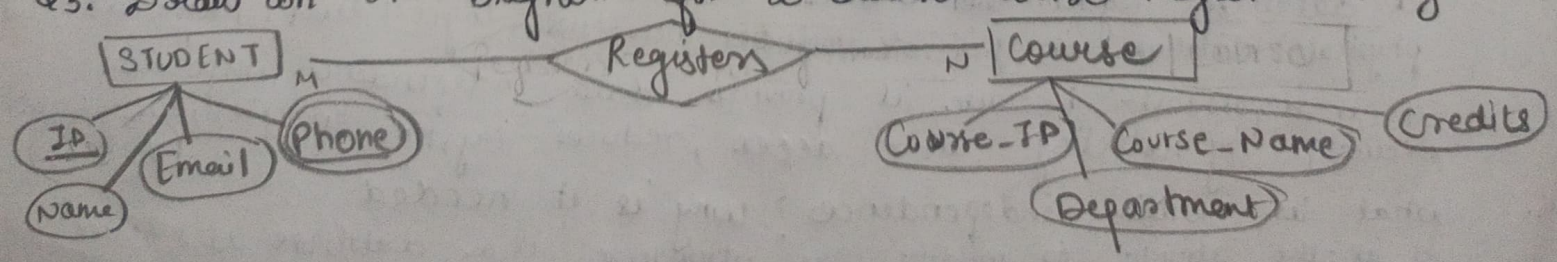
- The External Schema shows only the required part of database to diff users & hides the rest of data
- The Conceptual schema describes the overall structure of the database, including tables, attributes & relations
- The Internal Schema explains how data is actually stored in memory, such as file structures & indexing methods.

Q2. Draw an ER diagram for a Library Management System



Entities - MEMBER, BOOK, AUTHOR
Attributes - member - ID, Name, Address, Phone
Book - Book_ID, Title, Author, Publisher
Author - Author_ID, Author_Name
Relationship - Member borrows Book
Book written by Author.

Q3. Draw an ER Diagram for a Student Course Registration System



SECTION C

A hospital wants to store patient, doctor & appointment data. Identify entities, attributes & relationship.

ENTITIES: Patient, Doctor, Appointment

ATTRIBUTES: Patient: Patient_ID, Name, Age, Gender, Phone
Doctor: Doctor_ID, Name, Specialization, Phone
Appointment: Appointment_ID, Date, Time, Status.

Relationship: A Patient books Appointment
A Doctor attends Appointment

Q6. An online shopping platform stores customer, product & order data. Suggest keys & constraints

Ans

CUSTOMER	PRODUCT	ORDER
• Customer_ID (PRIMARY)	• Product_ID (PRIMARY)	• Order_ID (Primary key)
• EMAIL (UNIQUE)	• Product_name	• Customer_ID (Foreign key-customer)
• Name	• Price (CHECK>0)	• Order_Date
• Phone	• Stock (CHECK≥0)	• Total_Amount (CHECK≥0)

CONSTRAINTS

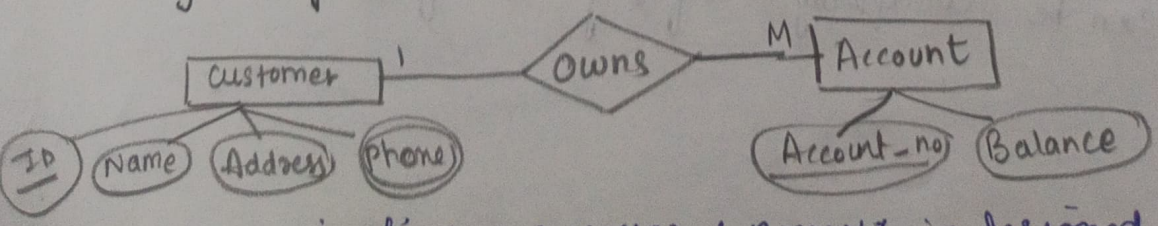
→ PRIMARY KEY CONSTRAINT: Ensures each record is unique (Customer_ID, Product_ID, Order_ID)
→ FOREIGN KEY CONSTRAINT: Maintains relationship b/w ORDER & CUSTOMER.
→ NOT NULL CONSTRAINT: Important fields like Name, Price, Order_Date can't be empty
→ UNIQUE CONSTRAINT: Email should be unique for each customer.
→ CHECK CONSTRAINT: Price and Stock must be +ve values.
→ REFERENTIAL INTEGRITY: An order cannot exist without a valid customer.

Q3. A university stores student records. Explain how data abstraction helps different users.

Ans In a university database system, data abstraction helps diff users access only the information they need without seeing the complete database details.
At the view level, students can see only their own records like marks attendance & personal details, while faculty can see marks & academic information of students they teach
At the logical level, administrators can view & manage the overall database structure such as student tables, course details & relationship
At the physical level, database administrators handle how the data is stored in files & memory, which is hidden from students & faculty

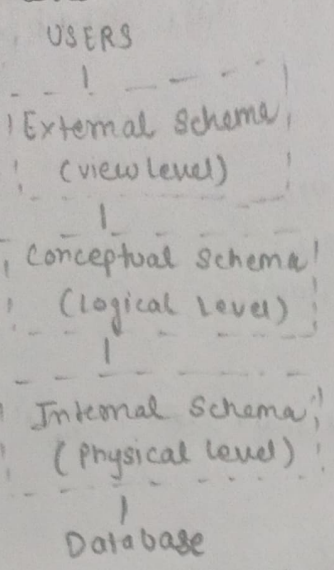
SECTION - D

Q1. Draw a simple ER diagram for any real-life system & explain one key design decision.



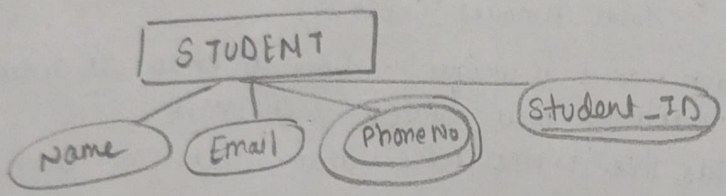
The relationship b/w CUSTOMER & ACCOUNT is designed as 1:M instead of 1:1 because a customer can have multiple accounts (saving, current) but each account is linked to only 1 customer.
This design makes the system flexible & realistic

Q2 Draw



- It provides data independence, meaning data changes at one level (like physical storage changes) do not affect the other levels or user application.

Q3. Draw an entity with attributes and identify its primary key.



Entity: Student
Attributes: Name, Email, Phone
Primary key: Student_ID

SECTION E

Q1. Write 2 assumptions you made while drawing an ER diagram

Ans
- Each entity has a unique primary key to identify every record uniquely.
- The relationship b/w entities follow real world rules (such as 1:M or M:M) based on system requirements.

Q2 Mention 1 limitation of ER modeling.

Ans One limitation of ER modeling is that it can't easily represent complex constraints or business rules (such as advanced calculation or procedural logic) beyond basic relationships & attributes.

Q3. What part of DBMS UNIT-1 did you find most difficult & why?

Ans I found data independence & the 3 schema architecture most difficult because understanding the diff b/w physical & logical levels & how changes at one level do not affect others can be conceptually confusing at first.